

Algorithmic Stability for Post-Hoc Selection

Nicholas Mino¹, Wenbin Zhou¹, and Shixiang Zhu¹

¹Carnegie Mellon University

Abstract

TBA

1 Introduction

[Wenbin]: The problem we study falls in the intersection of post-hoc inference, algorithm stability, and optimization theory.

2 Problem Setup

We observe data, run a data dependent selection algorithm, and then want to make an inference/evaluation statement about the selected object. The problem is that selection uses the same data as inference, so the selected object is statistically dependent on the data used to assess it.

We want to reduce or avoid data splitting by finding a general framework for post-hoc selection algorithms to convert a selection algorithm into a stable selection algorithm. Such a framework would seek to identify when a selection rule can be randomized or otherwise controlled in another way such that its output is sufficiently and certifiably stable for valid downstream inference or evaluation.

Let $D = (Z_1, \dots, Z_n)$ denote the observed dataset, where

$$Z_1, \dots, Z_n \stackrel{\text{i.i.d.}}{\sim} P,$$

and P is an unknown population distribution.

Let A be a data-dependent selection algorithm, and let

$$\hat{S} = A(D)$$

denote the selected object.

We seek a stabilized selection rule $\tilde{A}$ producing

$$\tilde{S} = \tilde{A}(D; \xi),$$

where $\tilde{A}$ is the stabilized selection algorithm, $\tilde{S}$ is the stabilized selected object, and ξ represents additional randomness or another stabilizing device. Depending on the stabilization mechanism, ξ may represent noise, subsampling randomness, randomized optimization perturbation, privacy randomness, or another source of algorithmic randomization.

The target post-selection inference guarantee is

$$\mathbb{P}(\theta_{\tilde{S}} \in C_{\tilde{S}}(D)) \geq 1 - \alpha$$

In words, after using the data to select an object, the confidence set reported for that selected object should contain its true parameter with probability at least $1 - \alpha$

Define more formally, including θ_S Define the policy as the solution to an optimization problem Policy: take input (context variable) ,... Formally define the stability of the policy induced by the optimization problem find stabilization mechanism

Application: ask any AI system what is the prediction of a certain quantity: "I want to know what the stock price is for apple tomorrow"

Think of an application: se

[Wenbin]: The post-hoc inference problem is itself very general. We need to narrow the structure of the problem by focusing on one specific objective (e.g., uncertainty quantification). See below for an example setting that I wrote.

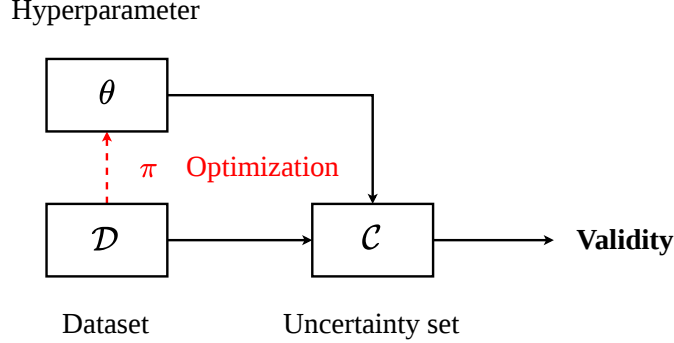


Figure 1: Illustration of the problem setting.

We consider a post hoc inference setting in uncertainty quantification problems, where the hyperparameter θ is determined via optimization using shared calibration data. Let $\mathcal{C}(X; \mathcal{D}, \alpha)$ denote an uncertainty set constructed with dataset $\mathcal{D}$ and input hyperparameter θ . Assume that the uncertainty set satisfies *validity*, i.e., for any given $\mathcal{D}$ and prespecified θ , there is

$$\mathbb{P}(Y \in \mathcal{C}(X; \mathcal{D}, \theta)) \geq 1 - \alpha. \quad (1)$$

Suppose there exist two functions $\hat{f}$ and $\hat{g}$ that are dependent on $\mathcal{D}^1$, and that they jointly form a constrained optimization problem that determines a hyperparameter θ :

$$\pi : \min_{\theta} \hat{f}(y; \theta) \quad \text{s.t.} \quad \hat{g}(\theta) \leq 0. \quad (2)$$

Let $\hat{\theta}(\mathcal{D})$ denote the output of (2). Our goal is to investigate: (i) the conditions when (1) still holds when $\hat{\theta}(\mathcal{D})$ is plugged in for θ and (ii) if not, what is the deviation between the nominal miscoverage and the actual miscoverage, and (iii) what kind of procedure can be deployed to recalibrate the uncertainty set to satisfy (1).

[Wenbin]: This objective is a mix of algorithm stability, post-hoc inference, and optimization theory. I believe that we should decompose the question into two main parts: first, study the stability of constrained optimizations (i.e., if the dataset changes by one data point, how much does the output of the optimization change? Do we need to make any structural assumptions on the optimization problems, like how $\hat{f}$ and $\hat{g}$ are learned from the data $\mathcal{D}$?). Second, study post-hoc inference on top of these results.

[Wenbin]: We may modify the problem setting along the way if we find there are more principal versions of it.

[Wenbin]: Data-driven policy (e.g., PTO), no recalibration, strategy: data-randomize to achieve validity guarantee, this answers (iii). Our paper: structured noise, depends on inverse optimization. Post-inference Tijana (Winner's Curse, Algorithm Stability).

¹Using filtration notations, $\sigma(\hat{f}) \subseteq \sigma(\mathcal{D})$ and $\sigma(\hat{g}) \subseteq \sigma(\mathcal{D})$.

3 Working Formalization: Stable Post-Hoc Inference for Optimization-Selected Policies

3.1 Introduction

We observe data, run a data dependent optimization-based selection algorithm, and then want to make an inference/evaluation statement about the selected object. The problem is that selection uses the same data as inference, so the selected object is statistically dependent on the data used to assess it.

We want to reduce or avoid data splitting by finding a general framework for post-hoc optimization-based selection algorithms to convert an optimization-based selection algorithm into a stable selection algorithm. Such a framework would seek to identify when a selection rule can be randomized or otherwise controlled in another way such that its output is sufficiently and certifiably stable for valid downstream inference or evaluation.

3.2 Common Setup

3.3 Part I: Stability of the constrained optimizations

3.3.1 Problem Formulation

3.4 Part II: Post-hoc inference after stable selection

3.4.1 Problem Formulation